

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[5 \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

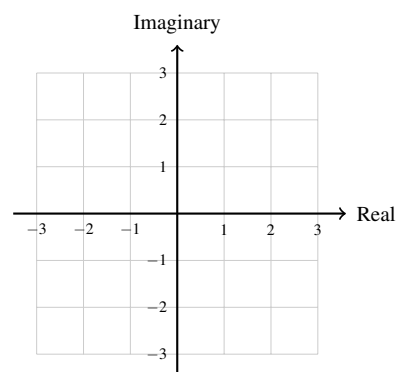
2. $\frac{\sqrt{7} + i\sqrt{21}}{-3\sqrt{21} - 3i\sqrt{7}}$

Find the power. Write the answer in rectangular form. (3 points)

3. $(-2 + 3i)^4$

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

4. $16 - 16i\sqrt{3}$



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

5. a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[5 \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = 2 \cdot 5 = 10 \quad \theta = \frac{5\pi}{4} + \frac{11\pi}{12} = \frac{15\pi}{12} + \frac{11\pi}{12} = \frac{26\pi}{12} = \frac{13\pi}{6}$$

$$\frac{13\pi}{6} - 2\pi = \frac{13\pi}{6} - \frac{12\pi}{6} = \frac{\pi}{6} \quad (\text{keep between } 0 \text{ and } 2\pi)$$

$$10 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 10 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \boxed{5\sqrt{3} + 5i}$$

Trap: given in radians, stay in radians. $2\pi = \frac{12\pi}{6}$, so subtract $\frac{12\pi}{6}$, not $\frac{2\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

Problem: $\frac{\sqrt{7} + i\sqrt{21}}{-3\sqrt{21} - 3i\sqrt{7}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

Top: $r = \sqrt{(\sqrt{7})^2 + (\sqrt{21})^2} = \sqrt{7 + 21} = \sqrt{28} = 2\sqrt{7}$ $\tan \theta = \frac{\sqrt{21}}{\sqrt{7}} = \sqrt{3} \Rightarrow \text{QI}, 60^\circ$

Bottom: $r = \sqrt{(-3\sqrt{21})^2 + (-3\sqrt{7})^2} = \sqrt{189 + 63} = \sqrt{252} = 6\sqrt{7}$ $\tan \theta = \frac{-3\sqrt{7}}{-3\sqrt{21}} = \frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIII} \Rightarrow 210^\circ$

$$\frac{2\sqrt{7} \operatorname{cis} 60^\circ}{6\sqrt{7} \operatorname{cis} 210^\circ} = \frac{1}{3} \operatorname{cis}(60^\circ - 210^\circ) = \frac{1}{3} \operatorname{cis}(-150^\circ) = \frac{1}{3} \operatorname{cis} 210^\circ$$

$$= \frac{1}{3} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \boxed{-\frac{\sqrt{3}}{6} - \frac{1}{6}i}$$

Trap: both parts negative means QIII. $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ says 30° ; the sketch says 210° . $\sqrt{252} = \sqrt{36 \cdot 7}$.

Find the power. Write the answer in rectangular form. (3 points)

Problem: $(-2 + 3i)^4$

3. De Moivre: $r^4 \operatorname{cis} 4\theta$. θ is not on the unit circle, so don't find it in degrees. Keep $\cos \theta = \frac{x}{r}$, $\sin \theta = \frac{y}{r}$ exact and use the double-angle formulas twice.

$$r = \sqrt{(-2)^2 + 3^2} = \sqrt{13}, \quad r^4 = 13^2 = 169 \quad \cos \theta = \frac{-2}{\sqrt{13}}, \quad \sin \theta = \frac{3}{\sqrt{13}} \quad (\text{QII})$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \frac{4-9}{13} = -\frac{5}{13} \quad \sin 2\theta = 2 \sin \theta \cos \theta = \frac{2(3)(-2)}{13} = -\frac{12}{13}$$

$$\cos 4\theta = \cos^2 2\theta - \sin^2 2\theta = \frac{25-144}{169} = -\frac{119}{169} \quad \sin 4\theta = 2 \sin 2\theta \cos 2\theta = \frac{2(-12)(-5)}{169} = \frac{120}{169}$$

$$169 \left(-\frac{119}{169} + \frac{120}{169}i \right) = \boxed{-119 + 120i}$$

Trap: $r^4 = (\sqrt{13})^4 = 169$, and it cancels every denominator. $2(-12)(-5)$ is positive. Check: squaring twice, $(-2 + 3i)^2 = -5 - 12i$, then $(-5 - 12i)^2 = -119 + 120i$.

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

Problem: $16 - 16i\sqrt{3}$

4. Convert, then roots. Radians throughout.

$$r = \sqrt{16^2 + (-16\sqrt{3})^2} = \sqrt{256 + 768} = \sqrt{1024} = 32$$

$$\tan \theta = \frac{-16\sqrt{3}}{16} = -\sqrt{3} \Rightarrow \text{ref } \frac{\pi}{3}, \text{ QIV} \Rightarrow \frac{5\pi}{3}$$

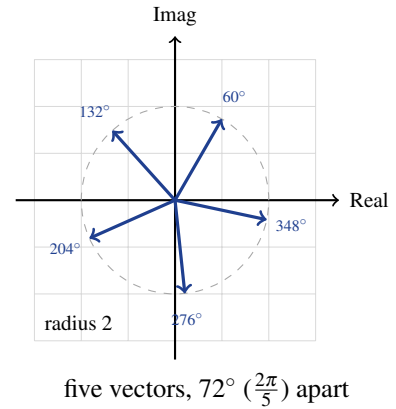
$$\sqrt[5]{32} = 2 \quad \text{first: } \frac{5\pi/3}{5} = \frac{\pi}{3} = \frac{5\pi}{15}$$

$$\text{spacing: } \frac{2\pi}{5} = \frac{6\pi}{15}$$

$$2 \text{cis } \frac{\pi}{3}, 2 \text{cis } \frac{11\pi}{15}, 2 \text{cis } \frac{17\pi}{15}, 2 \text{cis } \frac{23\pi}{15}, 2 \text{cis } \frac{29\pi}{15}$$

Common denominator 15 makes the $+\frac{6\pi}{15}$ steps easy. Five roots, last one $\frac{29\pi}{15} < 2\pi$.

Trap: spacing is $\frac{2\pi}{n}$, not $\frac{2\pi}{3}$ or $\frac{\pi}{5}$.



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

Problem: a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

5. All n th roots have the same r and sit $\frac{360^\circ}{n}$ apart. Find the given root's trig form, then step around.

$$-\sqrt{2} + i\sqrt{2}: r = \sqrt{2+2} = 2 \quad \tan \theta = -1, \text{ QII} \Rightarrow 135^\circ \quad \text{spacing } \frac{360^\circ}{4} = 90^\circ$$

$$\text{a) } 2 \text{cis } 225^\circ = 2 \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = -\sqrt{2} - i\sqrt{2}$$

$$2 \text{cis } 315^\circ = 2 \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \right) = \sqrt{2} - i\sqrt{2}$$

$$2 \text{cis } 405^\circ = 2 \text{cis } 45^\circ = 2 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = \sqrt{2} + i\sqrt{2}$$

$$\text{b) } w = (2 \text{cis } 135^\circ)^4 = 2^4 \text{cis } 540^\circ = 16 \text{cis } 180^\circ = 16(-1 + 0i) = \boxed{-16}$$

$$\text{a) } -\sqrt{2} - i\sqrt{2}, \sqrt{2} - i\sqrt{2}, \sqrt{2} + i\sqrt{2} \quad \text{b) } w = -16$$

Trap: fourth roots are 90° apart, not 120° (that is cube roots). To get w , raise the root to the 4th with De Moivre.